Experiment - 6.1.2. Factorial of a Number

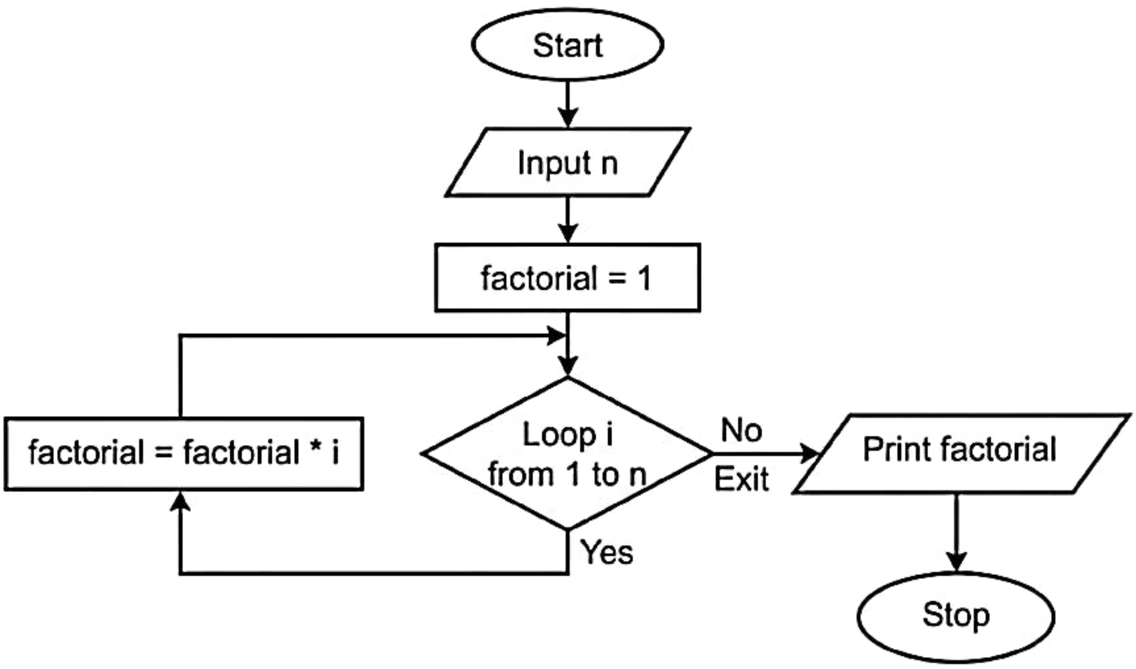
1. Aim

To design and implement a Python program that calculates the factorial of a given integer *n* using a for loop. The program reads the integer, computes its factorial iteratively, and prints the final result.

2. Pseudocode

1. **START**
2. **READ** a single integer from the user and **STORE** it in the variable n.
3. **INITIALIZE** a variable factorial to 1.
4. **FOR** each number i in the range from 1 to n (inclusive):
   - **MULTIPLY** factorial by i (factorial = factorial * i).
   - **STORE** the new value back into factorial.
5. **END FOR** loop once all numbers up to n have been multiplied.
6. **PRINT** the final value of factorial.
7. **END**

3. Flowchart



Programming and Problem Solving Lab - TE7287 - II Sem - 2026 Pin

Search course ctrl + k

6.1.2. Factorial of a Number

03:11

Write a Python program to calculate the factorial of a number n using loops.

Input Format:

- A single line containing an integer n.

Output Format:

- Print the factorial of the given integer n.

Sample Test Cases +

Explorer factorialN... Submit

```
n = int(input())
fact = 1
for i in range (1, n+1):
    fact = fact * i
print(fact)
```

Debugger

Average time	Maximum time	
0.003 s 3.00 ms	0.004 s 4.00 ms	2 out of 2 shown test case(s) passed 2 out of 2 hidden test case(s) passed

Test case 1 4 ms Debug

Expected output	Actual output
10	10
3628800	3628800

Test case 2 4 ms

Terminal Test cases